

Fprint DLL - developers guide

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Functions

OPEN_PORT			
Syntax	<pre>function OPEN_PORT(ComPort : byte; BaudRate : integer; StopBits : byte; Parity : byte; ByteSize : byte; DEVICE_INDEX : integer; Logical_number : integer; SERIAL_KEY : Widestring) : integer; stdcall;</pre>		
Description	Opens the serial port. Returns current status of the device, specified by the DEVICE_INDEX.		
Parameters	Parameter	Type	Description
	ComPort	byte	Defines serial port number
	BaudRate	integer	Defines port speed
	StopBits	byte	Serial port setting, usually use: StopBits = 0 Parity = 0 ByteSize = 8
	Parity	byte	
	ByteSize	byte	
	DEVICE_INDEX	integer	Index for the device - list of devices
	Logical_number	integer	Logical number of the device = 1
	SERIAL_KEY	Widestring	This is the key (given by our resellers when you get this DLL or FPrint) that corresponds to the serial number of your device.
Return values	Return codes		

OPEN_PORT_Ex

Syntax	function OPEN_PORT_Ex(ComPort : byte; BaudRate : integer; StopBits : byte; Parity : byte; ByteSize : byte; DEVICE_INDEX : integer; Logical_number : integer; SERIAL_KEY : PByteArray		
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CLOSE_PORT

Syntax	function CLOSE_PORT : integer;stdcall;
Description	Closes the serail port and destroys the current device object
Parameters	none
Return values	Return codes

OPEN_TCPIP

Syntax	function OPEN_TCPIP(IP_Address : WideString; TCP_Port : integer; DEVICE_INDEX : integer; SERIAL_KEY : WideString) : integer ; stdcall ;																	
Description	Opens a socket connection, defined by the IP and Port of the device. Returns current status of the device, specified by the DEVICE_INDEX.																	
Parameters	<table><tr><th>Parameter</th><th>Type</th><th>Description</th></tr><tr><td>IP_Address</td><td>Widestring</td><td>IP address of the device (eg. 192.168.0.1)</td></tr><tr><td>TCP_Port</td><td>integer</td><td>The port number of the device</td></tr><tr><td>DEVICE_INDEX</td><td>integer</td><td>Index for the device - list of devices</td></tr><tr><td>SERIAL_KEY</td><td>Widestring</td><td>This is the key (given by our resellers when you get this DLL or FPrint) that corresponds to the serial number of your device.</td></tr></table>	Parameter	Type	Description	IP_Address	Widestring	IP address of the device (eg. 192.168.0.1)	TCP_Port	integer	The port number of the device	DEVICE_INDEX	integer	Index for the device - list of devices	SERIAL_KEY	Widestring	This is the key (given by our resellers when you get this DLL or FPrint) that corresponds to the serial number of your device.		
Parameter	Type	Description																
IP_Address	Widestring	IP address of the device (eg. 192.168.0.1)																
TCP_Port	integer	The port number of the device																
DEVICE_INDEX	integer	Index for the device - list of devices																
SERIAL_KEY	Widestring	This is the key (given by our resellers when you get this DLL or FPrint) that corresponds to the serial number of your device.																
Return values	Return codes																	

OPEN_TCPIP_Ex

Syntax	function OPEN_TCPIP_Ex(IP_Address : WideString; TCP_Port : integer; DEVICE_INDEX : integer; SERIAL_KEY : PByteArray) : integer ; stdcall ;		
Description	Opens a socket connection, defined by the IP and Port of the device. Returns current status of the device, specified by the DEVICE_INDEX.		
Parameters	Parameter	Type	Description
	IP_Address	Widestring	IP address of the device (eg. 192.168.0.1)
	TCP_Port		The port number of the device
	DEVICE_INDEX	integer	Index for the device - list of devices
	SERIAL_KEY	PByteArray (max 15 bytes)	This is the key (given by our resellers when you get this DLL or FPrint) that corresponds to the serial number of your device.
Return values	Return codes		

CLOSE_TCPIP	
Syntax	function CLOSE_TCPIP : integer ;stdcall;
Description	Closes the socket connection and destroys current device object.
Parameters	none
Return values	Return codes
EXECUTE_STRING	
Syntax	function EXECUTE_STRING(var Execute_Value : Widestring) : integer ;stdcall;
Description	Sends one command to the device. The format of the strings and commands is described in the FPrint manual.
Parameters	Execute_Value : Widestring – the string to be executed - one command only. See the Fprint documentation for detailed description of the commands. E.g. - “P,1,_____,_,_;Text;” - this string prints Text The answer string on success will be something like: 0002,0002 – numbers returned from the device
Return values	Return codes

EXECUTE_STRING_Ex	
Syntax	function EXECUTE_STRING_Ex(Execute_Value : PByteArray) : integer ;stdcall;
Description	Sends one command to the device. The format of the strings and commands is described in the FPrint manual.
Parameters	Execute_Value : PByteArray – the string to be executed - one command only. Maximum 220 bytes. See the Fprint documentation for detailed description of the commands.
Return values	Return codes

EXECUTE_FILE

Syntax	function EXECUTE_FILE(Execute_file_name, Answer_file_name : WideString; Dos_text : boolean; ClassicAnswer : boolean) : integer ; stdcall ;		
Description	Executes the commands within a file (Fprint format) and saves the answer for each command in answer file. See the Fprint documentation for detailed description of the commands.		
Parameters	Parameter	Type	Description
	Execute_file_name	Widestring	The file to be executed
	Answer_file_name	Widestring	Answer file (if you dont pass file name, the answer is saved in the execute_file_name
	Dos_text	boolean	If true - converts chars (0x80 - 0xBF) into letters (0xC0 – 0xFF). Useful if you pass DOS generated strings to the device
	ClassicAnswer	boolean	Specifies the type of answer (status) in the answer file – see Fprint documentation; if true – Classic answer, if false – Error number
Return values	Return codes		

EXECUTE_FILE_Ex

Syntax	function EXECUTE_FILE(Execute_file_name, Answer_file_name : PByteArray; Dos_text : boolean; ClassicAnswer : boolean) : integer ;stdcall;		
Parameters	Parameter	Type	Description
	Execute_file_name	PByteArray	The file to be executed
	Answer_file_name	PByteArray	Answer file (if you dont pass file name, the answer is saved in the execute_file_name
	Dos_text	boolean	If true - converts chars (0x80 - 0xBF) into letters (0xC0 – 0xFF). Useful if you pass DOS generated strings to the device
	ClassicAnswer	boolean	Specifies the type of answer (status) in the answer file – see Fprint documentation; if true – Classic answer, if false – Error number
Return values	Return codes		

EXECUTE_FILE_UTF

Syntax	function EXECUTE_FILE(Execute_file_name, Answer_file_name : WideString; ClassicAnswer : boolean) : integer ; stdcall ;		
Description	Executes the commands within a file (Fprint format) and saves the answer for each command in answer file. The script file must be in UTF8 encoding.		
Parameters	Parameter	Type	Description
	Execute_file_name	Widestring	The file to be executed (UTF8)
	Answer_file_name	Widestring	Answer file (if you dont pass file name, the answer is saved in the execute_file_name
	ClassicAnswer	boolean	Specifies the type of answer (status) in the answer file – see Fprint documentation; if true – Classic answer, if false – Error number
Supported devices	Bulgarian devices only.		
Return values	Return codes		

EXECUTE_FILE_UTF_Ex

Syntax	function EXECUTE_FILE(Execute_file_name, Answer_file_name : PByteArray; ClassicAnswer : boolean) : integer ; stdcall ;		
Description	Executes the commands within a file (Fprint format) and saves the answer for each command in answer file. The script file must be in UTF8 encoding.		
Parameters	Parameter	Type	Description
	Execute_file_name	Widestring	The file to be executed (UTF8)
	Answer_file_name	Widestring	Answer file (if you dont pass file name, the answer is saved in the execute_file_name
	ClassicAnswer	boolean	Specifies the type of answer (status) in the answer file – see Fprint documentation; if true – Classic answer, if false – Error number
Supported devices	Bulgarian devices only.		
Return values	Return codes		

GET_SERIAL_NUMBER

Syntax	function GET_SERIAL_NUMBER(ComPort : byte; BaudRate : integer; StopBits : byte; Parity : byte; ByteSize : byte; DEVICE_INDEX : integer; Logical_number : integer; var SERIAL_NUMBER : WideString) : integer; stdcall;		
Description	Retrieves serial number for a device.		
Parameters	Parameter	Type	Description
	ComPort	byte	Defines serial port number
	BaudRate	integer	Defines port speed
	StopBits	byte	Serial port setting, usually use: StopBits = 0 Parity = 0 ByteSize = 8
	Parity	byte	
	ByteSize	byte	
	DEVICE_INDEX	integer	Index for the device - list of devices
	Logical_number	integer	Logical number of the device = 1
	SERIAL_NUMBER	Widestring	Output parameter for the serial number
Return values	Return codes		

GET_SERIAL_NUMBER_Ex

Syntax	function GET_SERIAL_NUMBER(ComPort : byte; BaudRate : integer; StopBits : byte; Parity : byte; ByteSize : byte; DEVICE_INDEX : integer; Logical_number : integer; var SERIAL_NUMBER : PByteArray) : integer; stdcall;		
Description	Retrieves serial number for a device.		
Parameters	Parameter	Type	Description
	ComPort	byte	Defines serial port number
	BaudRate	integer	Defines port speed
	StopBits	byte	Serial port setting, usually use: StopBits = 0 Parity = 0 ByteSize = 8
	Parity	byte	
	ByteSize	byte	
	DEVICE_INDEX	integer	Index for the device - list of devices
	Logical_number	integer	Logical number of the device = 1
	SERIAL_NUMBER	PByteArray (max 10 bytes)	Output parameter for the serial number
Return values	Return codes		

GET_SERIAL_NUMBER_TCPIP

Syntax	function GET_SERIAL_NUMBER_TCPIP(IP_Address : Widestring; TCP_Port : integer; DEVICE_INDEX : integer; var SERIAL_NUMBER : WideString) : integer ; stdcall ;		
Description	Retrieves serial number for a device, connected over TCP_IP.		
Parameters	Parameter	Type	Description
	IP_Address	Widestring	IP address of the device (eg. 192.168.0.1)
	TCP_Port		The port number of the device
	DEVICE_INDEX	integer	Index for the device - list of devices
	SERIAL_NUMBER	Widestring	Output parameter for the serial number
Return values	Return codes		

GET_LAST_ERROR

Syntax	function GET_LAST_ERROR : integer; stdcall
Description	Returns the result code of the last executed function
Parameters	none
Return values	Return codes

GET_LAST_ERROR_MESSAGE

Syntax	function GET_LAST_ERROR_MESSAGE : WideString; stdcall;
Description	Returns message, corresponding to the last result code
Parameters	none
Return values	Return codes

GET_LAST_ERROR_MESSAGE_Ex

Syntax	function GET_LAST_ERROR_MESSAGE : PByteArray; stdcall;
Description	Returns message, corresponding to the last result code
Parameters	none
Return values	Return codes

USE_OPERATOR			
Syntax	function USE_OPERATOR(operatorNumber : Integer; operatorPass: PByteArray; passLength : integer) : integer; stdcall;		
Description	Changes the default operator in use for commands that oper fiscal receipt automatically (commands “S” and “P”).		
Parameters	Parameter	Type	Description
	operatorNumber	Integer	Operator number
	operatorPass	PByteArray	Operator password
	passLength	Integer	Password length
Supported devices	WARNING – the command is currently in use for Bulgarian devices only.		
Return values	Return codes		

custom_Command_01

Syntax	function custom_Command_01(cmd : integer; input_Data : PbyteArray; inputData_ArrSize : integer; output_Data : PbyteArray; var outputData_ArrSize : integer; statusBytes : PbyteArray; var statusBytes_ArrSize : integer): integer; stdcall;																										
Description	Executes custom command, according to the firmware device description. The function input is command number, input data and input data size. The output returns data and status bytes from the device, as well as result code.																										
Parameters	<table><tr><th>Parameter</th><th>Type</th><th>Description</th></tr><tr><td>cmd</td><td>integer</td><td>Command number to be executed</td></tr><tr><td>input_Data</td><td>PByteArray (max size 220 bytes)</td><td>Pointer to byte array (max size 220 bytes), containing the data for the command</td></tr><tr><td>inputData_ArrSize</td><td>integer</td><td>Bytes count for array “input_Data”</td></tr><tr><td>output_Data</td><td>PByteArray (max size 220 bytes)</td><td>Pointer to byte array, containing the output data for the command from the device</td></tr><tr><td>outputData_ArrSize</td><td>integer</td><td>Bytes count for array “output_Data”</td></tr><tr><td>statusBytes</td><td>PByteArray (max size 8 bytes)</td><td>Pointer to byte array, containing the status bytes of the device after executing the command</td></tr><tr><td>statusBytes_ArrSize</td><td>integer</td><td>Bytes count for array “statusBytes”</td></tr></table>			Parameter	Type	Description	cmd	integer	Command number to be executed	input_Data	PByteArray (max size 220 bytes)	Pointer to byte array (max size 220 bytes), containing the data for the command	inputData_ArrSize	integer	Bytes count for array “input_Data”	output_Data	PByteArray (max size 220 bytes)	Pointer to byte array, containing the output data for the command from the device	outputData_ArrSize	integer	Bytes count for array “output_Data”	statusBytes	PByteArray (max size 8 bytes)	Pointer to byte array, containing the status bytes of the device after executing the command	statusBytes_ArrSize	integer	Bytes count for array “statusBytes”
Parameter	Type	Description																									
cmd	integer	Command number to be executed																									
input_Data	PByteArray (max size 220 bytes)	Pointer to byte array (max size 220 bytes), containing the data for the command																									
inputData_ArrSize	integer	Bytes count for array “input_Data”																									
output_Data	PByteArray (max size 220 bytes)	Pointer to byte array, containing the output data for the command from the device																									
outputData_ArrSize	integer	Bytes count for array “output_Data”																									
statusBytes	PByteArray (max size 8 bytes)	Pointer to byte array, containing the status bytes of the device after executing the command																									
statusBytes_ArrSize	integer	Bytes count for array “statusBytes”																									
Return values	Return codes																										
Supported devices	Bulgarian and Federation of Bosnia and Herzegovina “X” devices																										

custom_Command_02			
Syntax	function custom_Command_02(cmd : integer; statusBytes : PByteArray; var statusBytes_ArrSize : integer): integer ; stdcall ;		
Description	Executes custom command, according to the firmware device description. The function input is only command number, for commands that do not have input/output data. The output status bytes from the device, as well as result code.		
Parameters	Parameter	Type	Description
	cmd	integer	Command number to be executed
	statusBytes	PByteArray (max size 8 bytes)	Pointer to byte array, containing the status bytes of the device after executing the command
	statusBytes_ArrSize	integer	Bytes count for array "statusBytes"
Return values	Return codes		
Supported devices	Bulgarian and Federation of Bosnia and Herzegovina "X" devices		

custom_Command_03

Syntax	function custom_Command_03(cmd : integer; input_Data : PbyteArray; inputData_ArrSize : integer; output_Data : PbyteArray; var outputData_ArrSize : integer; statusBytes : PByteArray; var statusBytes_ArrSize : integer): integer ; stdcall ;		
Description	Executes custom command, according to the firmware device description. The function input is command number, input data and input data size. The output returns data and status bytes from the device, as well as result code.		
Parameters	Parameter	Type	Description
	cmd	integer	Command number to be executed
	input_Data	PByteArray (max size 220 bytes)	Pointer to byte array (max size 220 bytes), containing the data for the command
	inputData_ArrSize	integer	Bytes count for array “input_Data”
	output_Data	PByteArray (max size 220 bytes)	Pointer to byte array, containing the output data for the command from the device
	outputData_ArrSize	integer	Bytes count for array “output_Data”
	statusBytes	PByteArray (max size 8 bytes)	Pointer to byte array, containing the status bytes of the device after executing the command
	statusBytes_ArrSize	integer	Bytes count for array “statusBytes”
Return values	Return codes		
Supported devices	Bulgarian and Federation of Bosnia and Herzegovina “X” devices		

Regarding thread-safety

Fprint is not "Thread-safe"!

We do not manage the commands to the fiscal devices into a queue and if you send the commands to the fiscal device from multiple threads simultaneously – most probably some of the commands won't be executed correctly.

The fiscal device itself will reject some of them.

For example: If you want to sell simultaneously from multiple threads, but without managing your own queue – most probably the fiscal device will reject the second opening of the fiscal receipt.

The reason is the restriction by law. You have no right to open a second fiscal receipt if the first one is not closed properly.

FPrint works properly and sends commands to the fiscal device correctly – but you must to know about this peculiarity, related to the usage of the program from multiple threads simultaneously and to design your project according to that info.

Device indexes

INDEX	DEVICE	TCP/IP support
Montenegro		
2001	FP 550	No
2002	FP 1000	No
Ethiopia		
3001	FP 60	No
Federation of Bosnia and Herzegovina		
4001	FP 550	No
4002	TM-T260F	No
4003	FP 700 with RS232 communication interface	No
4004	FP 700 with TCP/IP communication interface	Yes
4005	FP 60 with RS232 communication interface	No
4006	FP 60 with TCP/IP communication interface	Yes
4007	FMP 350	No
4008	DP 25 X	Yes
4011	FP 700 X	Yes
4012	FP 60 X	Yes
4013	FP 700 MX	Yes
4100	MP 55 LD	No
4101	DP 50 Plus	No
4102	DP 55	No
4103	DP 55 S	No
4104	DP 25 MX	Yes
4105	DP 150 MX	No
4106	WP 50 MX	No
Serbia		
6001	FP 600 KL	No
6002	DP-05 KL	No
6003	DP-25 KL	No
6004	DP-35 KL	No
6005	DP-45 KL	No
Georgia		

7001	DP-25	No
7002	DP-05	No
Moldova		
8001	DP-25	Yes
8002	DP-150	No
8003	WP-500	Yes
8005	DP-25 MX	Yes
8006	DP-150 MX	No
8007	WP-50 MX	Yes
8008	FP-700 MX	Yes
Bulgaria		
10001	DP 05	No
10002	DP 25	No
10003	DP 35	No
10004	DP 150	No
10005	WP 50	No
10006	WP 500 X	Yes
10007	DP 25 X	Yes
10008	DP 150 X	No
10009	WP 50 X	No
10010	DP 05C X	No
10011	DP 15	No
10012	DP 05B	No
10013	DP 25 MX	Yes
10014	DP 150 MX	No
10015	WP 50 MX	No
11001	FP 650	No
11002	FP 800	Yes
11003	FP 2000	Yes
11004	FMP 350 X	No
11005	FP 700 X	Yes
11006	FMP 55 X	No
11007	FP 700	Yes
11008	FMP 10	No
11009	FP 700 MX	Yes

11010	FP 700 XE	Yes
11011	FP 700 XR	Yes

Error codes

VALUE	MEANING
0	
-1	General error / Unknown error
-2	Create error
-3	Open error
-4	Close error
-5	Device or resource busy
-6	Timeout expired
-7	The operation is not supported
-8	Memory allocation error
-9	Invalid parameter
-10	Input/Output error
-11	CRC error
-12	Flash memory error
-13	EEPROM error
-14	Device error
-15	The operation is not implemented
-16	The device or resource does not exists
-17	Invalid command
-18	Object does not exist
-19	No more items or operations
-20	Command failed
-21	Invalid command
-22	Not registered
-23	Error which was obtained in GetStatus method. Some bits in the status bytes of the fiscal device were raised
-24	Trial period has expired
-25	The functionality of your product is time limited. Please check for a new version or contact the support team!
- 40 till - 46	Device communication exceptions, check if the device is on.
-50	The printer is out of paper.
-51	General/syntax error – could be one of the following:

	- command returned error while executing; - command syntax error or invalid cmd number; - RAM failiure after power ON; - operational memory was cleared;
-52	Incorrect command number in the string
-53	Opened fiscal receipt, command not allowed
-54	Opened nonfiscal receipt, command not allowed
-55	No opened receipt, command not allowed
-56	Fiscal memory full
-57	Fiscal memory error (read/write failed/corrupt, memory in read-only state or last entry is corrupt)
-58	Command cannot be executed in current fiscal mode
-59	Ram reset after powering the device
-60	Incoming data has syntax error or wrong command number
-61	Failure in the printing mechanism
-62	Fiscal memory is in read only mode
-63	Tax terminal not responding
-64	Electronic journal is full
-201 till -220	Class exceptions due to wrong parameters/bugs
0	No errors or exceptions
-10000	Wrong logical number, check your string or file
-10001	Function "EXECUTE_FILE" - execute file does not exist
-10002	Function "EXECUTE_FILE" - wrong file format. Something is wrong in the string within the file – check Fprint documentation for help about the commands structure.
-10003	Function "OPEN_TCPIP" - wrong IP address string
-100000	Function "OPEN_PORT" – the serial key is wrong
-5000	Exception in "OPEN_PORT" function - check your parameters
-5001	Exception in "EXECUTE_STRING" function - check your parameters
-5002	Exception in "EXECUTE_FILE" function - check your parameters
-5003	Exception in "CLOSE_PORT" function
-5004	Exception in "GET_SERIAL_NUMBER" function - check your parameters
-5005	Exception in "GET_SERIAL_NUMBER_TCPIP" function - check your parameters
-5006	Exception in "OPEN_TCPIP" function - check your parameters
-5007	Exception in "CLOSE_TCPIP" function

Important: *All possible error codes returned from the devices are described in the file “Error_codes.ods”. You can find it into the “Help” folder.*

Examples

C# source code samples on how to use the functions can be found in the folder of the installation of FPrint:

..\Program Files (x86)\Datecs Applications\FPrintWIN\OpenSource\DLLDemo